ENGINEERING PORTFOLIO
2026

EMBEDDED · IOT · PCB · MANUFACTURING

Smart electronics, engineered end to end.

ATS Design & Innovation is a New Delhi-based product engineering company. We design, prototype and manufacture embedded, IoT and connected products — from PCB and firmware to cloud integration and OEM handover.

EMBEDDED FIRMWARE | PCB DESIGN | BLE / WI-FI / GSM | SENSORS | CLOUD & OTA | EMS

05
ENGINEERING CASE STUDIES

08
CORE SERVICE AREAS

15+
INTERFACES & PROTOCOLS

CONTACT

aayushrajchauhan8085@gmail.com
+91 88252 25508 · +91 62619 47121
B-71/1, 1st Floor, Mayapuri Industrial Area Phase 1,
New Delhi — 110064

REGISTRATIONS

MSME (Udyam): UDYAM-DL-03-0090955
GSTIN: 07BWQPC1160G1ZQ
Make in India · Atmanirbhar Bharat

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Appendix Company Profile

This portfolio presents ATS Design & Innovation's engineering capabilities, product lines and selected programme experience. Project notes distinguish development platforms from production hardware; exact part numbers and interfaces are confirmed against the target schematic and BOM.

01 / ABOUT ATS

Engineering depth. Manufacturing reality.

An electronics engineering partner built for the full product journey.

ATS Design & Innovation is a technology-driven company engaged in electronics product design, PCB development, embedded systems, IoT solutions and electronics manufacturing services (EMS). We build innovative, cost-effective and reliable electronic solutions for diverse industries, with a special focus on audio mantra products and smart bell solutions.

Our engineering work spans medical devices, marine monitoring, remote tracking, BLE sensor systems and cloud-connected IoT prototypes. Wherever the product sits, the discipline is the same: clear requirements, documented interfaces, verified hardware and firmware, and a clean handover.

From a first proof of concept to bulk production, we stay accountable for the details that decide whether a product performs in the field — not just in the lab.

COMPANY AT A GLANCE

COMPANY	ATS Design & Innovation
MSME (UDYAM)	UDYAM-DL-03-0090955
GSTIN	07BWQPC1160G1ZQ
LOCATION	New Delhi, India
FOCUS	Embedded · IoT · PCB · EMS
MANUFACTURING	Prototype to bulk production

MAKE IN INDIA · ATMANIRBHAR BHARAT · VOCAL FOR LOCAL · MSME

View company profile card

✓ Customized Solutions	✓ Quality & Reliability	✓ Cost-Effective Designs	✓ On-Time Delivery
✓ Technical Support	✓ Innovation Focused	✓ Long-Term Partnership	✓ Manufacturing Ready

INDUSTRIES & APPLICATIONS

Medical devices · Marine & remote monitoring · Tracking & telematics · BLE sensor systems · Smart security · Audio & devotional products

01 — Design
Requirements, architecture, electronics and PCB.

02 — Prototype
Bring-up, firmware integration and rapid iteration.

03 — Validate
Functional, connectivity, power and security testing.

04 — Manufacture
BOM, assembly, test and production support.

05 — Support
Field issues, improvements and lifecycle engineering.

We measure success by what works in the field — not by what looks good in a demo.

02 / SERVICES

One partner from circuit to shipment.

Our service lines are designed to work together — one accountable engineering partner instead of a chain of handoffs. We start with your requirements and carry the work through design, verification and production support.

01

Product Design & R&D

Concept definition, system architecture, electronics and mechanical co-design, feasibility studies and design-for-manufacture reviews.

02

Embedded System Development

Bare-metal and RTOS-based firmware on STM32, ESP32 and ARM platforms: peripheral drivers, boot/update flows, diagnostics and fault recovery.

03

IoT & Smart Product Solutions

BLE, Wi-Fi, GSM, GPS and MQTT connectivity; device provisioning, telemetry, remote control and cloud integration.

04

PCB Design & Support

Schematic capture, multilayer layout, BOM engineering, signal and power integrity checks, fabrication and assembly support.

05

Prototyping & Testing

Board bring-up, peripheral validation, connectivity testing, fault isolation and rapid design iteration on real hardware.

06

Electronics Manufacturing

PCB assembly, testing, rework and bulk production, backed by production-ready documentation and controlled build processes.

07

OEM / ODM & Custom Solutions

Customized electronics, private-label products and manufacturing support for long-term OEM and ODM partnerships.

08

Technical Support & Lifecycle

Prototype debugging, field issue investigation, design improvements and responsive engineering support after delivery.

03 / PRODUCTS & MANUFACTURING

Built, tested and produced in-house.

Real products and production capability — not just engineering concepts. Our lines include smart bells, mantra audio products, PCB assembly and custom electronics.



FOCUS AREA

Mantra Chanting & Bell Solutions

Spiritual audio products and bell solutions designed, assembled and supported by our team.

- Mantra chanting products manufacturing
- Custom audio mantra solutions
- Mantra chanting bell
- Door bell — wired / wireless
- Water overflow alarm bell
- Musical door bell
- Custom audio bell with a specific mantra or sound



SMART HOME

Smart Door Bells

Smart, wired and wireless door bell solutions with custom audio and reliable everyday performance.



DEVOTIONAL AUDIO

Mantra Bell

Spiritual and devotional products combining audio, lighting and considered industrial design.



PCB & ASSEMBLY

PCB Manufacturing

Prototype to production: fabrication, assembly and testing for reliable boards.



DESIGN

Electronics Design

PCB and IoT solutions from system architecture to verified hardware.



PRODUCT ENGINEERING

Product Development

From concept and enclosure to final product, with manufacturing in mind.



PRODUCTION

EMS & Assembly

SMT assembly, testing, rework and bulk production with controlled quality.

04 / SELECTED ENGINEERING WORK

Engineering case studies.

Five programmes spanning embedded vision, remote tracking, medical firmware, BLE sensor systems and cloud-connected IoT. Each entry reflects hands-on engineering work — the platform, the interfaces and the problems solved. Development platforms are distinguished from production hardware; exact part numbers are confirmed against the target schematic and BOM.

SELECTED WORK AT A GLANCE

01	02	03	04	05
Marine fish tracking	**Remote tracking**	**Insulin pump firmware**	**BLE provisioning**	**ESP32 cloud OTA**
Embedded vision · NPU · UART/RS-485	GSM · GPS · Iridium · BLE	STM32 · BLE · AES-CCM	Sensors · flash · OLED	Wi-Fi · MQTT · AWS S3

Each case study follows the same format: platform, interfaces, engineering work and relevance to a connected security product.

CASE STUDY 01

EMBEDDED VISION · EDGE AI · MARINE

Marine Fish Tracking & Underwater Monitoring

Embedded vision and marine monitoring development: capture underwater images, run image processing and AI inference, then send compact results to a companion communication device for delivery to a mobile application.

PLATFORM / HARDWARE

STM32N6570-DK evaluation platform (STM32N6 family), B-CAMS-IMX camera module, IMX335 image sensor, onboard NPU, external Delphi Nanomodem communication unit.

INTERFACES / PROTOCOLS

UART and RS-485 between embedded/companion equipment; camera sensor interface and STM32N6 image/NPU pipeline.

ENGINEERING WORK

- Camera bring-up and image capture pipeline investigation on the STM32N6 platform.
- AI model deployment exploration using STM32 AI tooling and ONNX/TFLite model workflows.
- NPU-based inference investigation, including fish detection and compact result representation.
- Image compression/encoding and compact data transfer design to reduce payload over the communication link.
- Defined payload fields for detection identity, fish type, size-related information and metadata; planned UART/RS-485 integration to the external modem.

RELEVANT TO Hub-side event processing, edge intelligence, serial communication with radio/modem subsystems, structured payload design and efficient transmission of event information.

Scope note: *The STM32N6570-DK is an evaluation platform used for development. Final production hardware, storage design and exact interface mapping depend on product requirements.*

CASE STUDY 02

REMOTE TRACKING | CELLULAR | SATELLITE

Marine / Remote Tracking & Monitoring

Firmware integration and system verification for a remote tracking platform combining location, cellular and satellite communications, BLE beacon detection and external depth/temperature sensor data.

PLATFORM / HARDWARE

STM32-based embedded controller and Cavli modem integration environment; GSM/cellular modem, GPS receiver, Iridium satellite module, BLE beacon interface, external depth & temperature sensor, configuration storage and debug/serial interfaces.

INTERFACES / PROTOCOLS

BLE, GSM, GPS, Iridium and UART/serial interfaces. Application-level configuration commands included DTADD, DTINT, DTRM, DT, DT?, II, ION and IRAL.

ENGINEERING WORK

- Implemented and verified configuration flows for BLE, GSM, GPS and satellite communication.
- BLE beacon detection and retrieval of depth/temperature data; verified automatic disconnect and reconnect when a sensor moved out of range and returned.
- Connection retry and recovery logic, including aborting a failed BLE connection attempt and restarting scanning.
- Verified GPS capture intervals, GSM transmission intervals/timeouts, and transmission of waypoints and events to the server.
- Tested store-and-forward, debug log retrieval and remote configuration; fixed configuration persistence and device identity issues, including synchronising modem IMEI information into configuration storage.
- System-level testing with staging-server configuration, signal-strength and modem-registration dependencies.

RELEVANT TO Supervising remote wireless sensors, detecting missing devices, handling reconnection, processing alarms/events, maintaining diagnostics and forwarding information through cellular or backup paths.

Scope note: *The programme included several hardware revisions and modem configurations. Exact MCU, modem model and pin-level peripheral assignments should be taken from the applicable schematic/BOM.*

CASE STUDY 03

MEDICAL | BLE | SECURITY

Insulin Pump Firmware & Security

Embedded firmware engineering in a regulated medical-device context, with emphasis on reliable MCU firmware, BLE connectivity, peripheral integration and security mechanisms.

PLATFORM / HARDWARE

STM32L496ZG / STM32L4 development platform; X-NUCLEO-BNRG2A1 BLE expansion board using BlueNRG-2; ST-LINK V3 for programming and debug.

INTERFACES / PROTOCOLS

Embedded C/C++, BLE, SPI, I2C, UART, ADC, DAC, timers/PWM; AES-CCM, SHA-256, HMAC-SHA256, PBKDF2-HMAC-SHA256.

ENGINEERING WORK

- Developed and debugged STM32 firmware using STM32CubeIDE and STM32 HAL/LL drivers.
- Integrated BlueNRG-2 BLE and mobile-side testing with ST BLE Toolbox; worked on pairing, provisioning, privacy-related behaviour and secure communication concepts.
- Implemented and integrated cryptographic building blocks with the STM32 cmox library, including AES-CCM, SHA-256 and HMAC operations; used PBKDF2 with SHA-256 for key derivation and authentication-related flows.
- Peripheral driver development and board-level integration for sensors and interfaces.
- Worked within a quality and compliance-oriented development environment, with exposure to IEC 62304, ISO 13485:2016 and IEC 60601-1-8 documentation expectations.

RELEVANT TO Secure device onboarding, encrypted wireless communication, firmware architecture, peripheral integration, traceable development practices and security-conscious product engineering.

Scope note: *Reflects embedded engineering contributions and technology exposure; it does not claim certification of a complete product or independent regulatory approval.*

CASE STUDY 04

BLE SENSORS PROVISIONING

BLE Device Provisioning & Sensor Integration

BLE-enabled device development and sensor interfacing across STM32 evaluation platforms, including provisioning, configuration and acquisition of environmental and system data.

PLATFORM / HARDWARE

STM32L496ZG on NUCLEO-L496ZG; BlueNRG-2 on X-NUCLEO-BNRG2A1; STM32WB55 and STM32U585 evaluation exposure; SHT40 temperature/humidity sensor, TMP107 temperature sensor, ADS1115 four-channel ADC, W25Q128FW / AT25Q128A external flash, NHD-1.69-160128UGC3 OLED display.

INTERFACES / PROTOCOLS

BLE, I2C, SPI, UART, ADC, GPIO and timer interfaces.

ENGINEERING WORK

- Integrated BLE modules with STM32 firmware and tested device discovery/provisioning workflows.
- Developed sensor interfacing for temperature, humidity and multi-channel ADC acquisition.
- Worked on external flash driver integration and non-volatile data handling.
- Integrated OLED display hardware and embedded display data paths.
- Performed peripheral debugging, communication validation and firmware bring-up.

RELEVANT TO Security sensor integration, local status display, device configuration, event logging and reliable communication between a hub and peripheral modules.

CASE STUDY 05

IOT CLOUD OTA

ESP32 Cloud OTA & Device Control

Connected-device prototypes exploring remote control, cloud-hosted firmware delivery and over-the-air update workflows.

PLATFORM / HARDWARE

ESP32-C6 development module and ESP32 camera-module development hardware; WS2812 addressable RGB LED; Wi-Fi; power and GPIO interfaces.

INTERFACES / PROTOCOLS

Wi-Fi, MQTT using PubSubClient, Blynk IoT platform, HTTP/HTTPS, AWS S3 presigned URLs and a Node.js/Express backend.

ENGINEERING WORK

- Built an IoT control prototype using Blynk and MQTT for remote RGB LED control; used MQTT Explorer for broker/device connectivity troubleshooting.
- Developed a Node.js/Express web service concept to list firmware binaries and initiate OTA updates.
- Integrated AWS S3 presigned URL workflows for controlled firmware download.
- Investigated OTA download failures, HTTP response errors and timeout behaviour.
- Worked with ESP32-C6 GPIO and WS2812 LED control.

RELEVANT TO Cloud-to-device workflows, remote configuration, OTA update orchestration, Wi-Fi connectivity and connected-device troubleshooting.

Scope note: *These are prototype and integration activities, not a claim of a production-scale cloud platform or a certified OTA security implementation.*

APPLICATION TO YOUR PRODUCT

What this experience brings to a connected security platform.

01

Hub firmware & interfaces

Embedded development on ARM and ESP32 platforms, peripheral integration, diagnostics and communication management.

02

Wireless sensors

BLE provisioning, device supervision, reconnect behaviour and sensor integration across real hardware.

03

Wide-area connectivity

GSM, GPS, satellite and Wi-Fi/MQTT experience for primary and backup communication paths.

04

Security & cloud

Cryptographic implementation exposure, secure BLE concepts and OTA/cloud update workflows.

05 / TECHNICAL STACK

Platforms, protocols and tools.

A practical stack built on STM32 and ESP32 platforms, with the interfaces, security primitives and cloud workflows needed for connected products. Platform choices are always confirmed against the target hardware and requirements.

MCU / COMPUTE

- STM32L496ZG / STM32L4
- STM32N6570-DK / STM32N6
- STM32WB55
- STM32U585
- ESP32-C6
- ARM-based platforms

TOOLS & FIRMWARE

- STM32CubeIDE
- STM32CubeMX
- STM32CubeProgrammer
- ST-LINK V3
- TouchGFX
- X-CUBE-BLE
- STM32 AI / ST Edge AI
- ESP-IDF / Arduino
- VS Code
- Embedded C/C++
- STM32 HAL/LL
- FreeRTOS exposure
- Boot / update workflows
- Diagnostics

CONNECTIVITY

- BLE / BlueNRG-2
- GSM / cellular
- GPS
- Iridium satellite
- Wi-Fi
- MQTT
- HTTP/HTTPS
- UART
- RS-485

PERIPHERALS & INTERFACES

- SPI
- I2C
- UART
- ADC
- DAC
- GPIO
- Timers / PWM
- External flash
- OLED
- Environmental sensors
- Camera sensor
- NPU pipeline

SECURITY

- AES-CCM
- SHA-256
- HMAC-SHA256
- PBKDF2-HMAC-SHA256
- BLE security / privacy concepts
- Secure firmware design exposure

CLOUD & APPLICATION

- AWS S3 presigned firmware delivery
- Node.js / Express
- MQTT
- Blynk
- Server-bound telemetry
- Mobile-side BLE testing

Platform selections are always confirmed against the target schematic, BOM and product requirements.

LOW-POWER SENSING

STM32L4 / STM32WB55

BLE sensor nodes, provisioning, external flash, local display and battery-conscious firmware for distributed devices.

REMOTE & CELLULAR

STM32 + GSM / GPS / Iridium

Store-and-forward telemetry, reconnect logic, diagnostics and backup communication paths for remote assets.

EDGE AI & VISION

STM32N6 + camera / NPU

Image capture, on-device inference and compact event payloads for bandwidth-constrained links.

We select the simplest platform that meets the requirement, then verify it on real hardware before committing to production design.

06 / FEATURED ENGAGEMENT BLUEPRINT

Smart Security Hub.

Our proposed approach for a customized wireless security ecosystem built around an intelligent hub — coordinating connected sensors and alarm devices, managing local events, communicating with cloud services and providing secure control through Android and iOS applications. This is a proposed engagement blueprint, not completed work.

PROPOSED SYSTEM ARCHITECTURE

01 / EDGE
Security devices
Door/window contacts, PIR sensors, smoke/gas detectors, panic buttons, sirens and tamper inputs.

→

02 / LOCAL INTELLIGENCE
Intelligent hub
ARM-based compute, embedded Linux, local event engine, device supervision, storage, diagnostics and power management.

→

03 / CLOUD
Cloud services
Device and user management, event ingestion, remote diagnostics, notification orchestration and OTA coordination.

→

04 / EXPERIENCE
Mobile applications
Account access, installation workflow, pairing, arm/disarm, live status, alerts, event history and permissions.

Zigbee / Thread / dedicated RF · BLE commissioning · Wi-Fi · Ethernet · LTE · Secure APIs · OTA updates

PROPOSED WORK PACKAGES

Six workstreams, one integrated system.

A — Product definition & architecture
Confirm use cases, device count, radio protocols, operating environment, power targets, security model, regulatory markets and production constraints. Produce a system block diagram, interface definition and development plan.

B — Hardware & PCB engineering
Evaluate ARM compute platforms and connectivity modules; define memory and storage; design power input, battery charging and backup; integrate wired and wireless interfaces; prepare schematics, PCB layout, BOM and prototype test plan.

C — Embedded Linux & hub firmware
Establish board support and boot flow; configure the Linux image and required services; implement hardware interfaces, device communication, provisioning, event handling, local diagnostics, logging and update mechanisms.

D — Security-device integration
Define onboarding and supervision flows for sensors and actuators; implement pairing, device identity, health monitoring, tamper reporting, alarm prioritisation, lost-link handling and recovery behaviour.

E — Mobile & cloud integration
Define secure APIs and event flows for user accounts, hub registration, remote control, notifications, history and device management. Finalise mobile and backend scope with the client.

F — Security, verification & handover
Develop a threat-informed security plan covering identity, authenticated communication, secure boot/update capabilities where supported, key handling, access control and data protection. Execute functional and integration tests; deliver source, design files and manufacturing documentation.

1 — Discovery
Requirements, use cases, constraints, system architecture and risk register.

2 — Proof of concept
Platform and radio evaluation; early hub-to-device communication demonstration.

3 — Engineering prototypes
PCB prototypes, Linux/firmware baseline, device integration and initial app/cloud interfaces.

4 — Integrated system
End-to-end alarm flows, remote control, notifications, diagnostics and security hardening.

5 — Verification & handover
System testing, issue closure, manufacturing package, source code and technical documentation.

ⓘ **Recommended first step:** a paid discovery and architecture phase, followed by milestone-based prototype development and system validation. This reduces platform and integration risk before committing to the full production design.

07 / DELIVERY & HANDOVER

Yours to own, build and extend.

Good engineering is more than working hardware. It is documented interfaces, reproducible builds and a package your team can own, maintain and manufacture.

Handover-ready by design

The expected package for an agreed scope typically includes:

- System architecture, interface specifications and configuration records.
- Schematics, PCB source/layout files, Gerbers, BOM and assembly notes, where included in scope.
- Firmware and application source code, build instructions, dependency versions and release notes.
- Linux configuration/build materials and deployment instructions for the selected platform.
- API definitions, cloud deployment documentation and database/interface descriptions, as applicable.
- Test plans, test reports, known-issues register and prototype verification results.
- Manufacturing and service documentation, programming instructions and handover support.

How we work

A disciplined, transparent process from the first conversation to production.

- **Requirements before code.** We clarify the product, constraints and acceptance criteria before committing to a design.
- **Risk reduction first.** Platform and radio feasibility is validated before full development begins.
- **Milestone-based delivery.** Each phase has clear outputs, so progress is visible and reviewable.
- **Documented interfaces.** Version-controlled source, reproducible builds and release notes are part of the work.
- **Long-term partnership.** Support continues beyond prototype delivery, through production and field issues.

VERIFICATION & QUALITY

Validated before it leaves the bench.

Functional & integration
End-to-end device, hub, cloud and application flows.

Connectivity & range
Pairing, radio behaviour, reconnect and recovery.

Power & resilience
Startup, shutdown, fault and backup behaviour.

Security review
Identity, authenticated communication and update paths.

08 / CONTACT

Let's build something reliable.

Share your product requirements, target market and timeline. We'll respond with a clear technical approach, scope and next steps.

EMAIL
aayushrajchauhan8085@gmail.com

PHONE
+91 88252 25508 · +91 62619 47121

LOCATION
B-71/1, 1st Floor, Mayapuri Industrial Area Phase 1, New Delhi, South West Delhi, Delhi — 110064

Company details

Registrations and contact information for procurement and onboarding.

COMPANY
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Email ATS →

Open to new product development engagements

01 **Share the brief**
Tell us the product, target market, timeline and any platform constraints.

02 **Technical review**
We assess feasibility, risks and the right architecture for the scope.

03 **Scope & proposal**
You receive a clear plan, deliverables and next steps — no black boxes.

APPENDIX / COMPANY PROFILE

ATS at a glance.

The original company profile card — registrations, service lines, product portfolio and contact details.



ATS

DESIGN & INNOVATION

Turning ideas into smart electronic solutions. Product design, embedded systems, IoT and electronics manufacturing from New Delhi, India.

EXPLORE	CONTACT	REGISTRATIONS
About ATS	aayushrajchauhan8085@gmail.com	MSME (Udyam): UDYAM-DL-03-0090955
Services	+91 88252 25508	GSTIN: 07BWQPC1160G1ZQ
Products	+91 62619 47121	Make in India · Vocal for Local
Engineering work	New Delhi, India	
Approach		

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Engineering portfolio · Project descriptions reflect engineering contributions and technology exposure; scope-dependent details are confirmed per engagement.